NAME : KUMAR MILIND
REG : 23MIS0650
AGILE MIDTERM LAB

GitHub :- https://github.com/milinx/23mis0650_MidTerm_Agile

https://github.com/milinx

Q2 B>

2. Simple Static Web page.

TASK : CI and CD

- Write the code in Visual Studio and push it to your GITHUB repository.
- Modify the code and commit the updates back in GITHUB.
- Using Jenkins configure the job as pipeline and create the stages for the CI and CD process.
- Containerize the application using Dockerfile and display the stages in stage view in Jenkins.
- Create kubernetes cluster and visualize the deployment in the given node port.

a. The student results

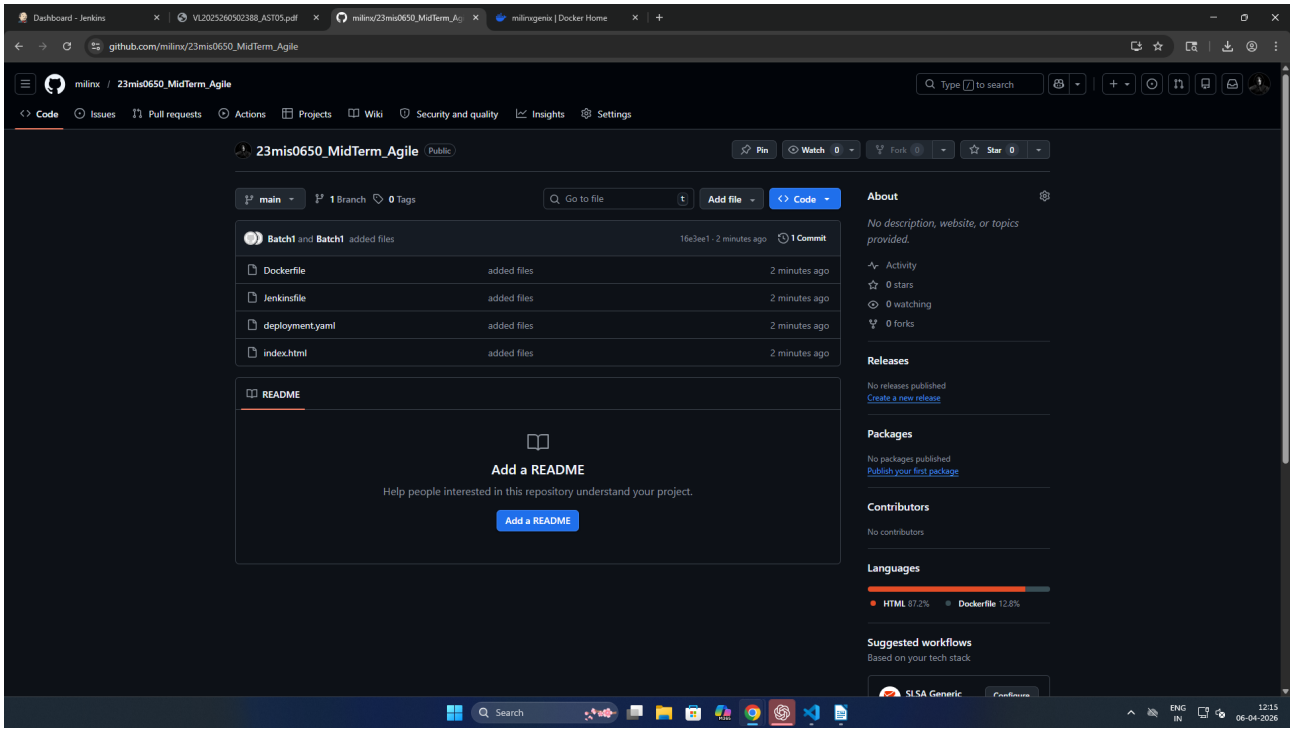
b. Create some five events during Rivera.

c. The Conference information and its full details.

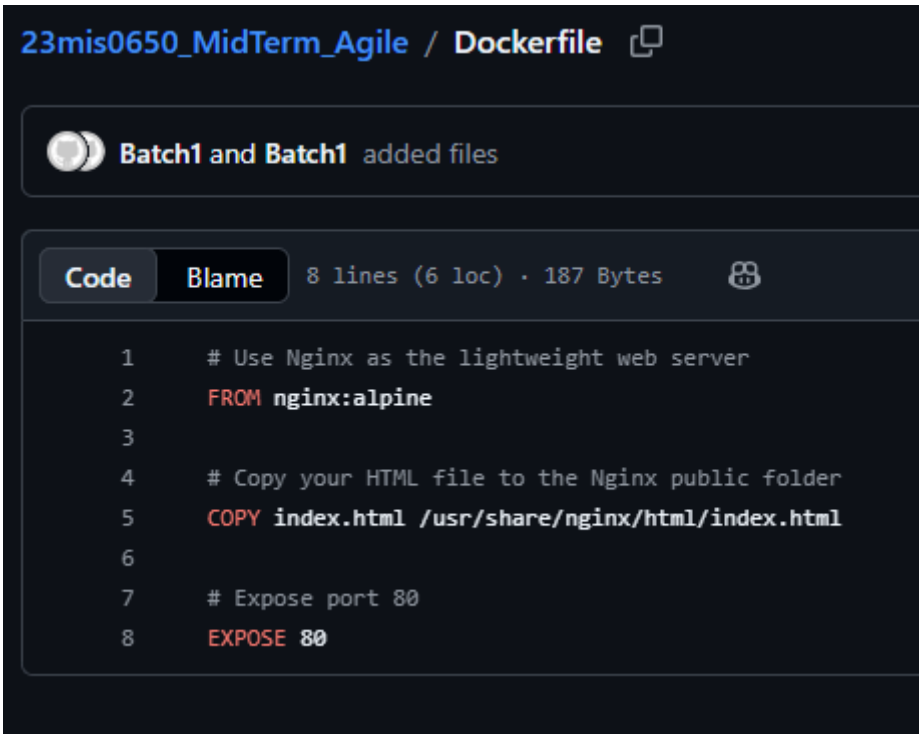
d. The price list of the items

e. The timetable of a student.

Created a code folder and files in vscode and pushed on GITHUB



Dockerfile



Jenkinsfile

23mis0650_MidTerm_Agile / Jenkinsfile

Batch1 and Batch1 added files

Code Blame 20 lines (20 loc) · 424 Bytes

```
pipeline {
    agent any
    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }
        stage('Build Docker Image') {
            steps {
                sh 'docker build -t rivera-app:latest .'
            }
        }
        stage('Deploy to Kubernetes') {
            steps {
                sh 'kubectl apply -f deployment.yaml'
            }
        }
    }
}
```

deployment.yaml

Files

main

Go to file

- Dockerfile
- Jenkinsfile
- deployment.yaml
- index.html

23mis0650_MidTerm_Agile / deployment.yaml

Batch1 and Batch1 added files

Code Blame 33 lines (32 loc) · 564 Bytes

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: rivera-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: rivera
  template:
    metadata:
      labels:
        app: rivera
    spec:
      containers:
      - name: rivera-container
        image: rivera-app:latest
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: rivera-service
spec:
  type: NodePort
  selector:
    app: rivera
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007 # Access the site at http://<node-ip>:30007
```

index.html

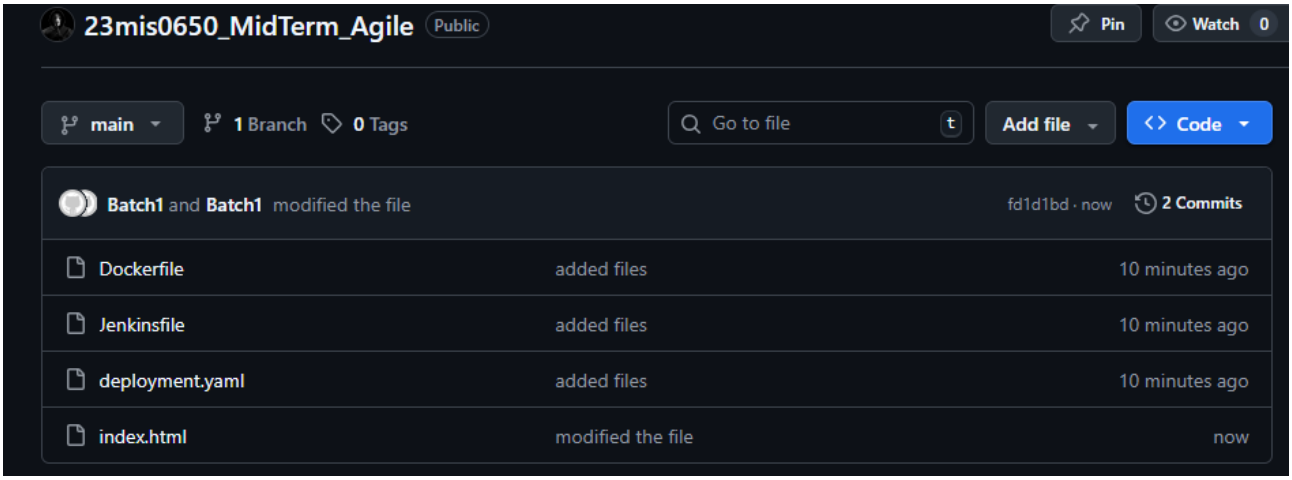
23mis0650_MidTerm_Agile / index.html

Batch1 and Batch1 added files

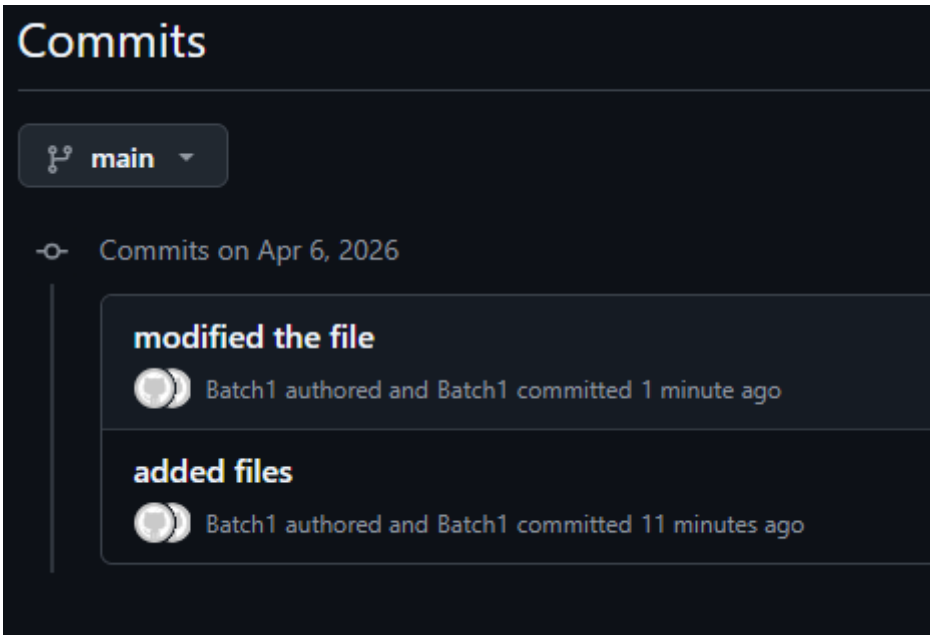
Code Blame 28 lines (28 loc) · 1.24 KB

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Rivera Events</title>
    <style>
        body { font-family: 'Segoe UI', sans-serif; background-color: #f0f2f5; display: flex; justify-content: center; padding: 50px; }
        .card { background: white; padding: 30px; border-radius: 12px; box-shadow: 0 4px 15px rgba(0,0,0,0.1); width: 400px; }
        h1 { color: #2c3e50; border-bottom: 2px solid #3498db; padding-bottom: 10px; }
        ul { list-style: none; padding: 0; }
        li { padding: 12px 0; border-bottom: 1px solid #eee; color: #34495e; }
        li:last-child { border-bottom: none; }
        strong { color: #e74c3c; }
    </style>
</head>
<body>
    <div class="card">
        <h1>Rivera Events</h1>
        <ul>
            <li><strong>1. Pro-Show:</strong> Live concert featuring celebrity artists.</li>
            <li><strong>2. Battle of Bands:</strong> A high-energy rock music competition.</li>
            <li><strong>3. Dance-Off:</strong> Inter-collegiate group dance face-off.</li>
            <li><strong>4. Game-a-thon:</strong> 24-hour e-sports and gaming tournament.</li>
            <li><strong>5. Art Gallery:</strong> A showcase of fine arts and photography.</li>
        </ul>
    </div>
</body>
</html>
```

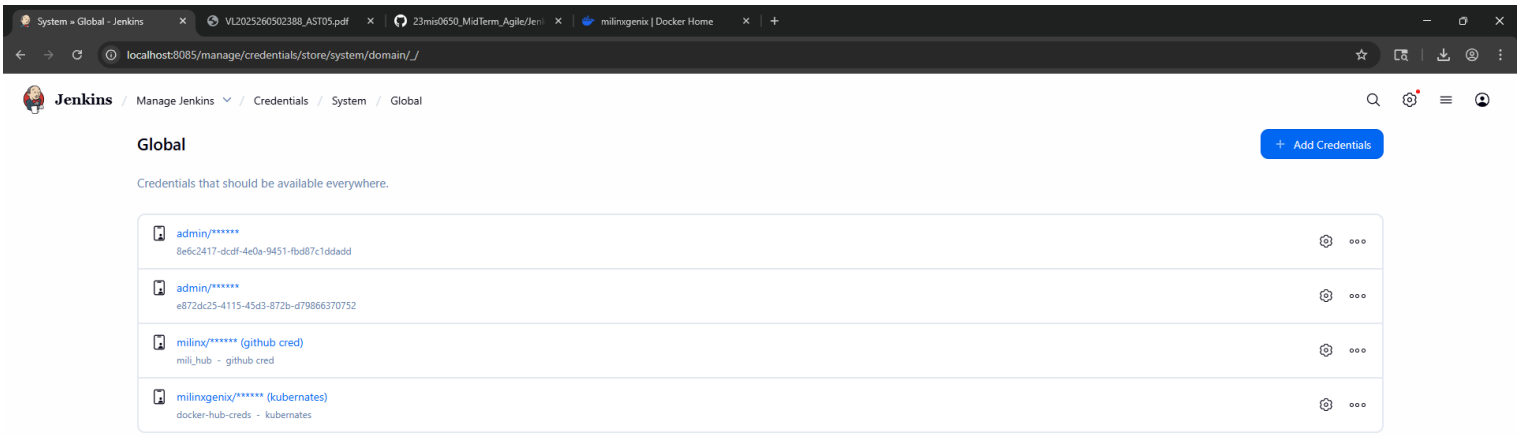
Modify the code and commit the updates back in GITHUB.



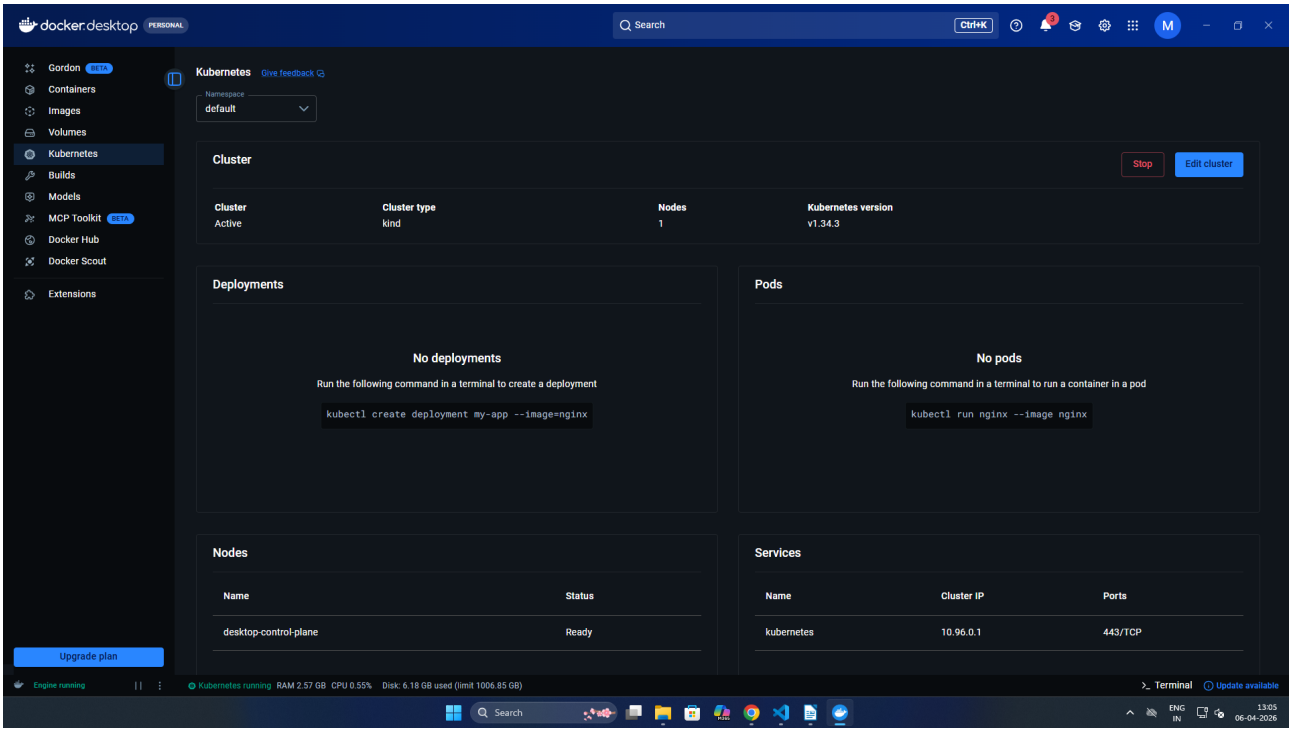
Modified index file and pushed the update on github



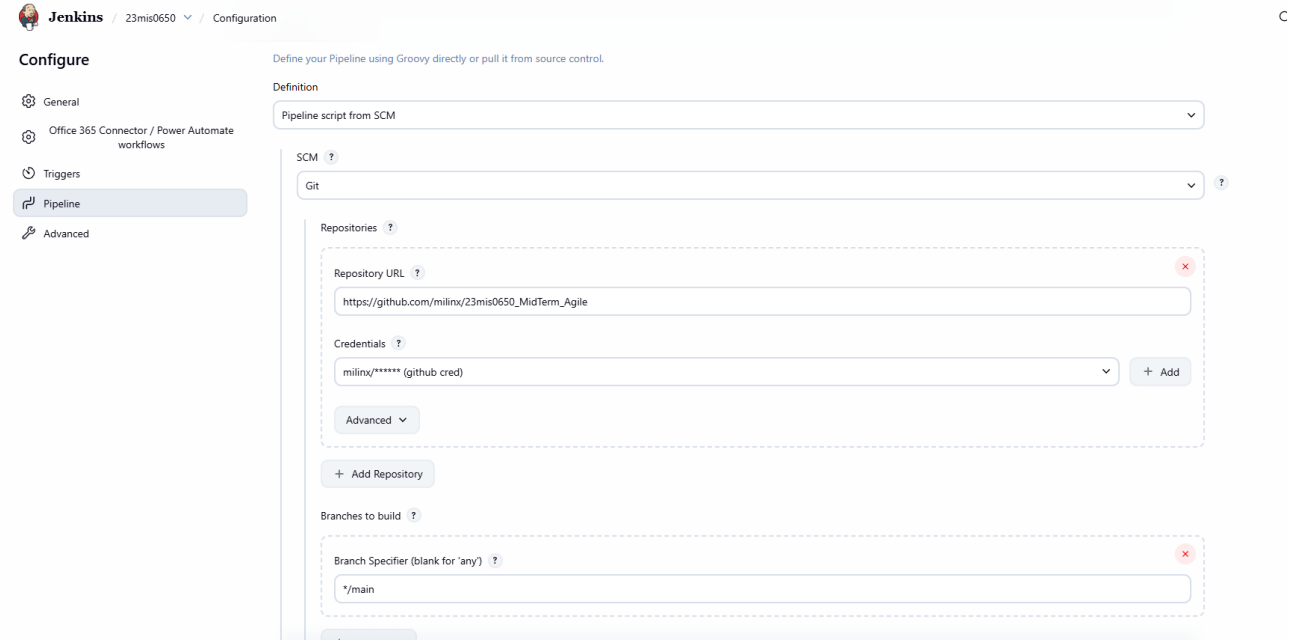
Added credentials in Jenkins to connect GitHub and Docker Kubernetes



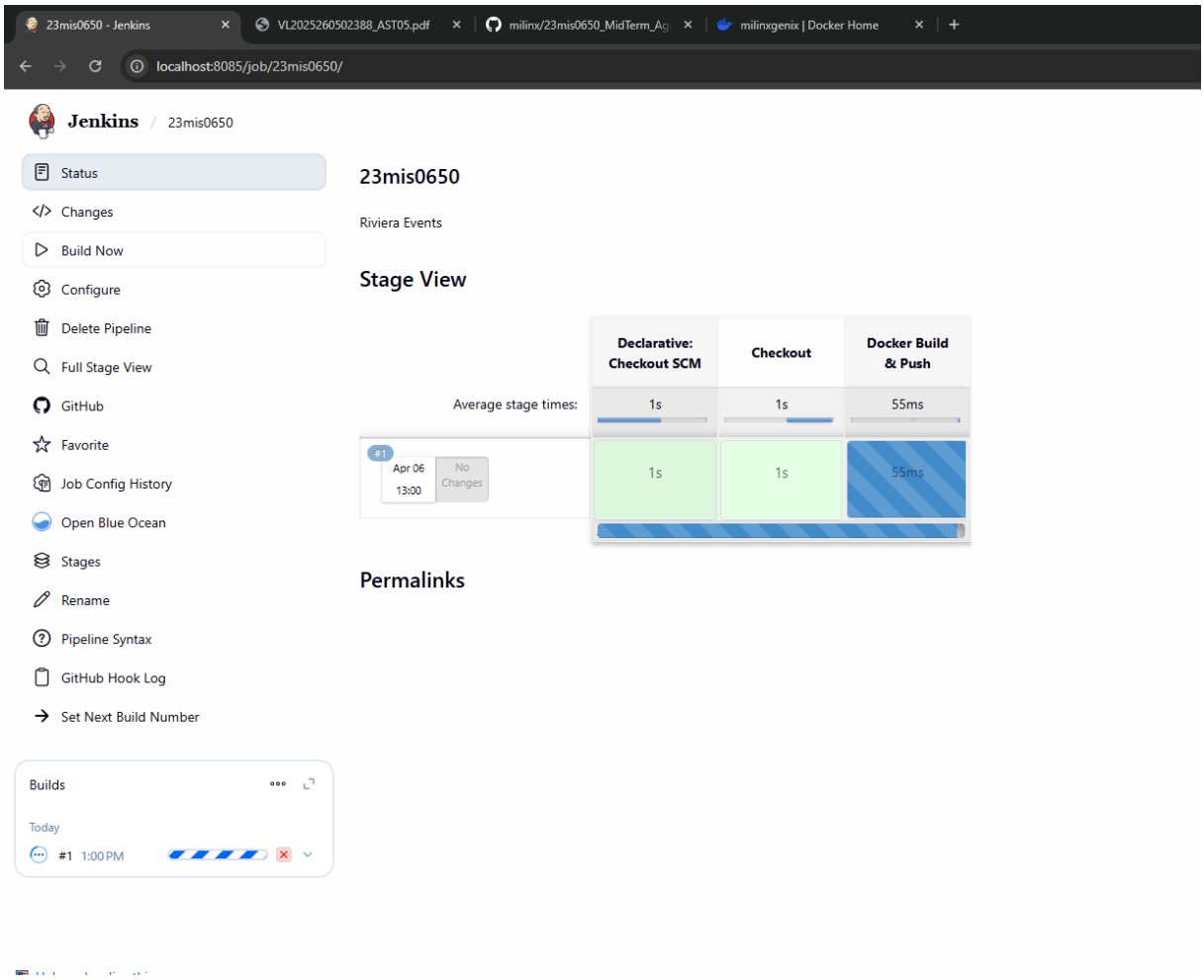
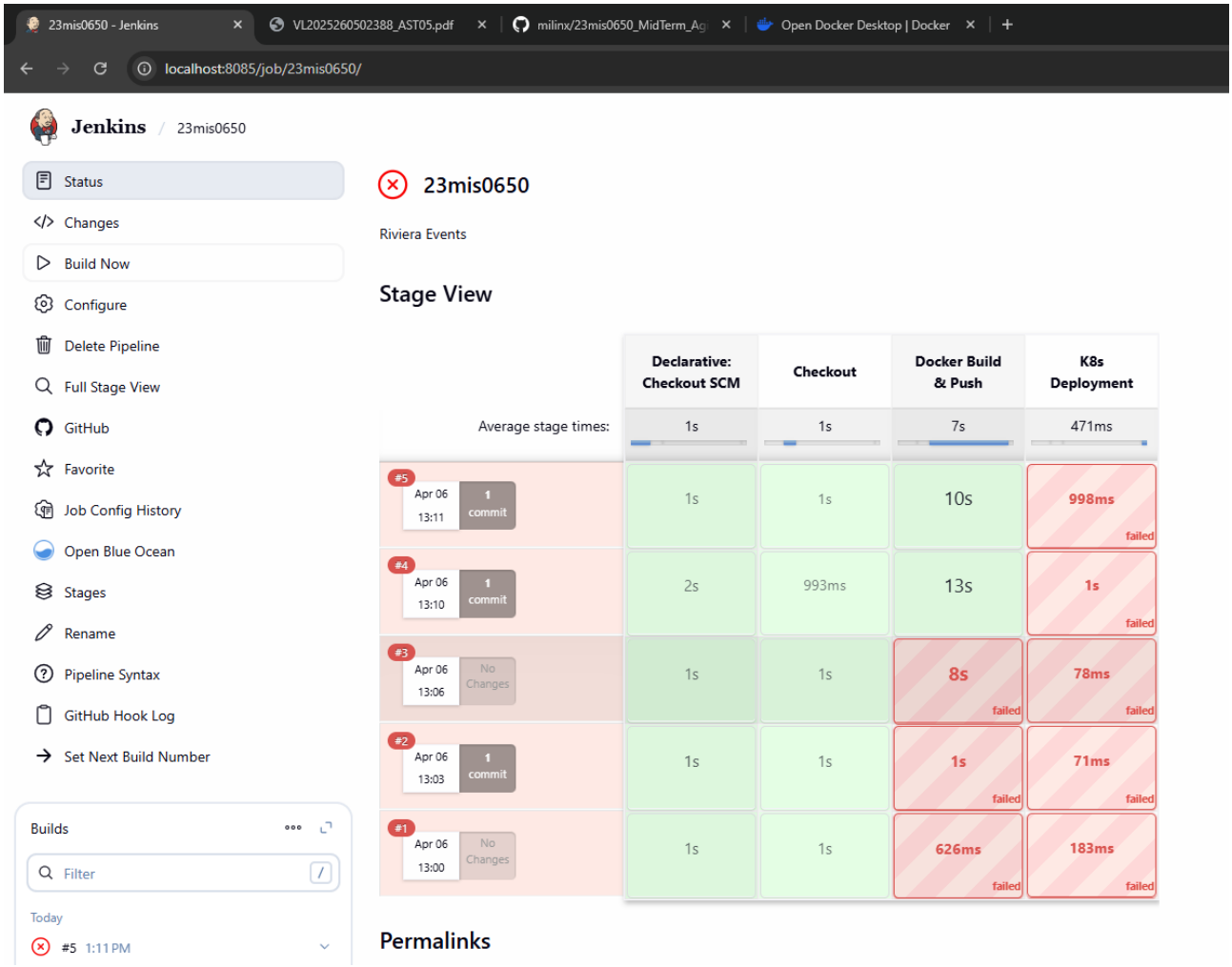
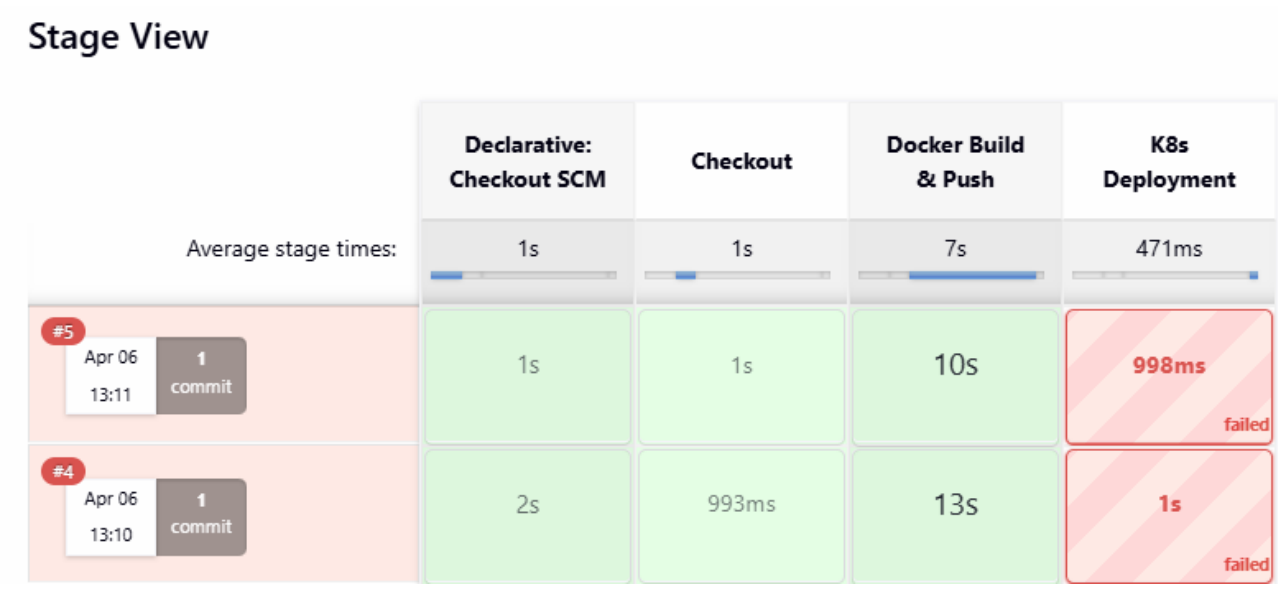
KUBERNETES



Jenkins Pipeline configuration



Building the project on Jenkins



4 stages passed stucking in last 1 stage